Short-Term Mental-Health Risk Forecasting Using Wearable, Sleep, Activity, Behavioral, and Self-Report Signals

Final Project Report

Applied Forecasting Methods

Dataset Used:	0017 jang ts.tsv
Forecasting Target:	Next-observed anxiety score
Practical Interpretation:	Mostly next-day anxiety forecasting
Input Window:	Previous 10 observations
Best Continuous Forecasting Model:	XGBoost
Main Evaluation Split:	70% chronological train, 30% chronological test

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1. Abstract

This project focuses on short-term mental-health risk forecasting using participant-level time-series data. The selected dataset, 0017_jang_ts.tsv, contains wearable, sleep, activity, behavioral, and self-report variables collected over time from multiple participants. The forecasting target is the participant's next observed anxiety score. Since 82% of the test samples are exactly one day apart, the forecasting task is practically close to next-day anxiety prediction.

The forecasting problem is formulated as a supervised time-series task. For each participant, records are sorted chronologically. The previous 10 observations are used as input to predict the next observed anxiety score. We compare baseline models, statistical time-series models, machine-learning models, and deep-learning models. The main continuous prediction metrics are MAE, RMSE, and R^2. For high-risk warning, anxiety score ≥ 2 is treated as a high-risk case, and alerting performance is evaluated using precision, recall, and F1-score.

The best continuous forecasting model is XGBoost. On the full next-observation test set, XGBoost achieves RMSE ≈ 0.584 and $R^2 \approx 0.627$. On the true next-day subset, XGBoost remains strong with RMSE ≈ 0.584 and $R^2 \approx 0.604$. This shows that the model's performance is not only due to vague next-observation prediction; it remains effective on actual next-day cases.

The project also shows that the best score predictor is not always the best warning model. XGBoost gives the strongest continuous prediction, while persistence-based rules and calibrated thresholds are important for high-risk anxiety alerting. Therefore, the final system should be interpreted as an early-warning support tool, not as a clinical diagnosis system.

2. Problem Statement

Current wearable and self-tracking systems usually provide reactive feedback. They can show current heart rate, sleep duration, activity level, or current mood, but they do not always forecast near-future mental-health risk. A reactive system may only warn the user after stress or anxiety has already increased.

The forecasting challenge is different. Instead of only asking:

> What is the user's current state?

we ask:

> Given the user's recent physiological, behavioral, sleep, activity, and self-report history, can we forecast their next anxiety level?

This is useful because a short-term forecast can support early warning. If a system can identify increasing anxiety risk before the next check-in, it can help the user take preventive actions such as rest, sleep correction, reduced workload, or seeking support.

The final project objective is:

> **To build and compare forecasting models that predict short-term future anxiety risk using recent participant history.**

The project does not claim to diagnose anxiety disorder, burnout, panic attacks, or clinical mental-health conditions. Anxiety score is used as a measurable mental-health risk indicator.

3. Project Goals

The project has four main goals.

3.1. Goal 1: Prepare a Valid Time-Series Forecasting Dataset

The dataset must be converted into a supervised forecasting format while preserving time order. For each participant, records are sorted by date, and the target is created from the next observed anxiety score.

Previous 10 observations → Next observed anxiety

3.2. Goal 2: Check Whether the Forecasting Horizon is Meaningful

Because the data is not perfectly continuous for every participant, the target is formally called next-observed anxiety. However, we calculate the gap between the current observation and the target observation. This allows us to check whether the task is close to next-day forecasting.

3.3. Goal 3: Compare Different Model Families

A forecasting project should not depend on only one model. We compare:

- baseline models,
- statistical time-series models,
- machine-learning models,
- deep-learning models.

3.4. Goal 4: Evaluate Both Prediction and Warning

A model can be good at predicting the average score but still poor at identifying high-risk cases. Therefore, we evaluate both:

1. continuous score prediction,
2. high-risk anxiety alerting.

4. Dataset Description

The final dataset used in this project is:

0017 jang ts.tsv

It is a participant-level longitudinal dataset. Each row corresponds to a dated observation for a participant. The dataset contains physiological, activity, sleep, lifestyle, and self-report variables.

4.1. Raw Dataset Summary

The raw dataset contains 3969 rows and 45 columns. There are 43 unique participants before filtering. The date range is from 14 May 2021 to 16 December 2023.

Table 1: Raw Dataset Summary

Property	**Value**
Rows	3969
Columns	45
Unique participants	43
Date range start	14 May 2021
Date range end	16 December 2023
Missing values	0
Duplicate rows	0
Duplicate id-date pairs	0

The absence of missing values and duplicate id-date pairs makes this dataset cleaner than the alternative datasets considered during the project.

4.2. Important Columns

The important columns include:

- id: participant identifier,
- date: date of observation,
- heart-rate variables such as hr_mean, hr_max, hr_var,
- step variables such as steps_mean, steps_maximum,
- sleep variable such as sleep_duration,
- lifestyle variables such as coffee, alcohol, and exercise,
- self-report variables such as anxiety, positive_feeling, negative_feeling, positive energy, and negative_energy.

4.3. Participant-Level Observation Counts

The number of observations per participant is not uniform. Some participants have long histories, while others have very few observations.

Table 2: Observation Count Summary per Participant

Statistic	**Value**
Number of participants	43
Mean observations per participant	92.30
Standard deviation	86.25
Minimum observations	1
25th percentile	24.50
Median observations	65.00
75th percentile	144.00
Maximum observations	327

This participant imbalance is important. Participants with very few records cannot provide enough history for a 10-observation forecasting window. Therefore, after sequence creation and filtering, the final supervised dataset contained 28 usable participants.

4.4. Top Participants by Number of Records

Table 3: Top Participants by Number of Observations

Participant ID	Number of Rows
SYM2-1-353	327
SYM2-1-285	292
SYM2-1-325	290
SYM2-1-412	234
SYM2-1-573	218
SYM2-1-405	182
SYM2-1-337	173
SYM2-1-563	161
SYM2-1-357	154
SYM2-1-422	154

These participants contribute most of the useful time-series information. Longer participant histories are more useful for sequence-based forecasting.

4.5. Low-Cardinality Variables

Some variables are binary or low-cardinality. These include lifestyle and event indicators.

Table 4: Low-Cardinality Variables

Column	Number of Unique Values
beep	1
alcohol	2
menstruation	2
coffee	2
panic	2
exercise	2
smoking	2
negative_feeling	4
positive_energy	4
negative_energy	4
annoying	4
anxiety	4
positive_feeling	4

The column beep is constant and therefore does not provide predictive information. Binary variables such as coffee, alcohol, exercise, and smoking may still be useful as behavioral indicators.

4.6. Final Supervised Dataset Summary

After cleaning, filtering, and sequence generation, the final supervised dataset summary is:

Table 5: Final Supervised Dataset Summary

Quantity	Value
Final users after filtering	28
Total supervised sequences	3458
Training sequences	2408
Testing sequences	1050
Window size	10 observations
Features per timestep	16
Flattened ML features	160
High-risk anxiety threshold	anxiety ≥ 2

4.7. Is the dataset balanced or not?

The target variable is the next-observed anxiety score, which takes integer values from 0 to 3. The distribution across the training set is not balanced. Lower anxiety scores (0 and 1) appear substantially more frequently than higher-risk scores (2 and 3). In the test set, only 19.24% of samples are classified as high-risk (anxiety ≥ 2), meaning approximately 81% of cases represent normal or low anxiety. This imbalance is important for two reasons. First, a model that always predicts low anxiety would achieve high accuracy but would be useless for early warning. Second, evaluation metrics such as recall and F1-score are more informative than raw accuracy for assessing high-risk detection performance. The imbalance also justifies the use of threshold tuning for classification-based alerting, as the default decision boundary of 0.5 may not be optimal when the positive class is rare.

5. Why This Dataset Was Selected

Before finalizing the dataset, multiple datasets were examined. The final decision was to use 0017 jang_ts.tsv [redacted] because it provided the strongest balance of forecasting validity, feature richness, and model performance.

Table 6: Comparison of Candidate Datasets

Dataset	Strength	Main Issue	Decision
s2_proper continuous.C	clsea[redacted]n physiological signals and continuous data	Target stress score appeared to be engineered from the same physiological variables used as predictors, creating possible target leakage	Rejected
general _ema.csv	Direct stress target and many users	Very high stress missingness and irregular next-observation gaps; only about 5.45% of test cases were exactly next-day	Used for comparison, not final
0017_jang_ts.tsv	Wearable, sleep, activity, behavioral, and self-report features; mostly daily; stronger forecasting results	Target is anxiety rather than direct stress	Selected as final dataset

The final dataset was selected because:

1. it has a clear future target: next observed anxiety,
2. it avoids the direct formula-based leakage issue found in the stress-score dataset,
3. it has richer predictors than the EMA-only stress dataset,
4. most test examples are true next-day cases,
5. it gives stronger model performance.

6. Forecasting Problem Definition

For each participant i, records are sorted chronologically:

$$(x_{i,1}, y_{i,1}), (x_{i,2}, y_{i,2}), \ldots, (x_{i,T_i}, y_{i,T_i})$$

where $y_{i,t}$ is the anxiety score at observation t.
The target is defined as:

$$y_{i,t+1} = \text{next observed anxiety score}$$

The input is the previous 10 observations:

$$X_{i,t-9:t} = [x_{i,t-9}, x_{i,t-8}, \ldots, x_{i,t}]$$

The forecasting task is:

$$X_{i,t-9:t} \rightarrow y_{i,t+1}$$

Since most observations are daily, this is practically:

$$\text{previous 10 days} \rightarrow \text{next-day anxiety}$$

However, some participants have missing days. Therefore, the exact formal target is called next-observed anxiety.

7. Next Observation vs Next Day

7.1. Next-Observed Anxiety

Next-observed anxiety means the next available anxiety record for the same participant. If the next record occurs tomorrow, it is next-day. If the participant skipped some days, the next observation may be after 2 or more days.

7.2. True Next-Day Anxiety

True next-day anxiety means that the target record is exactly one calendar day after the current observation:

$$\text{day gap} = 1$$

7.3. Why This Check Was Important

This check was important because a forecasting task is stronger if the forecast horizon is clear. If the data has irregular gaps, next-observation forecasting may become variable-horizon forecasting. Therefore, we calculated the day-gap distribution.

Table 7: Next-Observation vs True Next-Day Test Samples

Quantity	**Value**
Total next-observation test samples	1050
True next-day test samples	861
Percentage true next-day	82.0%
High-risk rate in all next-observation test samples	19.24%
High-risk rate in true next-day subset	18.23%

This shows that the next-observed anxiety task is mostly a next-day forecasting task. Therefore, the final work reports both:

1. full next-observation forecasting,
2. true next-day subset validation.

8. Data Engineering Pipeline

The data engineering pipeline converts the raw longitudinal dataset into a supervised forecasting dataset.

8.1. Step 1: Sort by Participant and Date

For each participant, all rows are sorted by date:

$$\text{sort by (id, date)}$$

This is necessary because time-series forecasting must preserve chronological order.

8.2. Step 2: Select Features

The final feature set includes 16 features per timestep.

Table 8: Final Feature Categories

Feature Category	**Features**
Heart-rate features	hr_mean, hr_var, hr_max, hr_mesor, hr_amplitude
Activity features	steps_mean, steps_maximum
Sleep feature	sleep_duration
Lifestyle features	alcohol, coffee, exercise
Self-report features	anxiety, positive_feeling, negative_feeling, positive_energy, negative_energy

8.3. Step 3: Create Future Target

The future target is created using a participant-wise shift:

$$\text{target}_{i,t} = \text{anxiety}_{i,t+1}$$

This means current and past features are used to predict future anxiety.

8.4. Step 4: Calculate Day Gap

The day gap is calculated as:

$$\text{day gap} = \text{target date} - \text{current input end date}$$

This helps separate true next-day samples from general next-observation samples.

8.5. Step 5: Create Rolling Windows

For each participant, rolling windows of 10 previous observations are created.

For deep-learning models:

$$X \in \mathrm{R}^{10\times16}$$

For traditional machine-learning models, the window is flattened:

$$10 \times 16 = 160$$

8.6. Step 6: Chronological Train-Test Split

For each participant:

$$\text{first 70\% records} \rightarrow \text{training}$$

$$\text{last 30\% records} \rightarrow \text{testing}$$

This avoids using future records to train predictions for past records.

9. Why 70% Training and 30% Testing?

An 80/20 split is common, but we used 70/30 because the test set must be large enough for reliable evaluation, especially for high-risk anxiety detection.

High-risk anxiety cases are not extremely frequent. If the test set is too small, recall and F1-score can become unstable. A 70/30 split gives:

$$2408 \text{ training sequences}$$

$$1050 \text{ testing sequences}$$

This gives enough data for training while preserving a sufficiently large future test set.

The split is chronological within each participant, so the evaluation simulates forecasting future records of participants whose earlier history is available.

10. Exploratory Time-Series Analysis

Before applying forecasting models, we performed exploratory analysis to understand the structure of the dataset, the target distribution, participant-level variation, time gaps, and possible temporal patterns.

10.1. Target Variable: Anxiety

The target variable for the final project is future anxiety. The current anxiety column has four unique values, from 0 to 3. For high-risk warning, we define high risk to be 1 if anxiety is greater than or equal to 2 and 0 otherwise. This threshold separates normal or low anxiety from elevated anxiety risk.

10.2. Time-Series Plot

A time-series plot was used to visualize how anxiety changes over time. The plot shows that anxiety is not a smooth periodic signal. It contains low-anxiety periods, sudden spikes, and participant-specific changes.

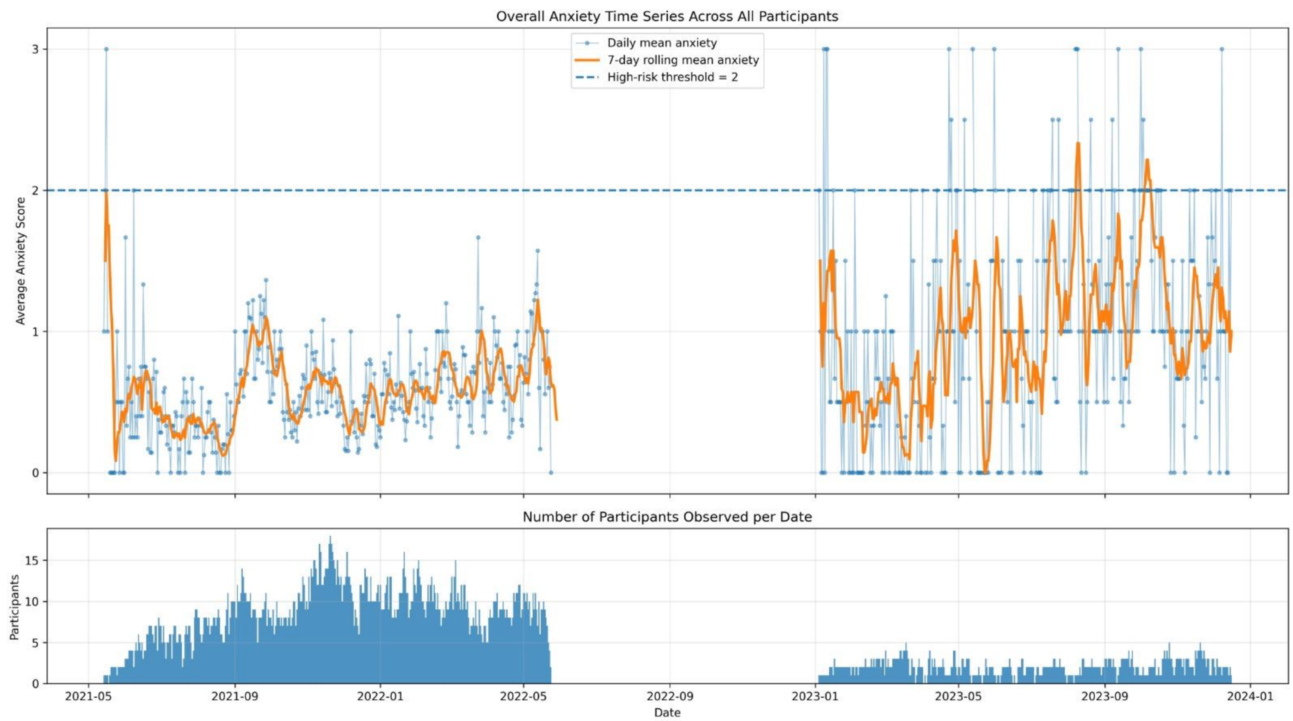

Figure 1: Anxiety time series over time. The target shows short-term persistence but also sudden spikes, making forecasting challenging.

10.3. Day-Gap Distribution

The day gap is the number of days between the current input end date and the target date. This is important because it tells us whether next-observed forecasting is close to next-day forecasting.

Table 9: Day-Gap Distribution in Test Set

Day Gap	Count
1	861
2	77
3	28

4	14
5	8
6	10
7	6
8	3
9	6
10	9
11	6
13	1
14	1
15	2
16	2
17	1
18	1
19	2
21	1
22	1
26	1
27	1
30	1
35	1
36	1
39	1
41	1
45	1
58	1
65	1

Out of 1050 test samples, 861 are exactly next-day samples. Therefore:

$$\frac{861}{1050} \times 100 = 82.0\%$$

This confirms that the next-observed task is mostly next-day forecasting.

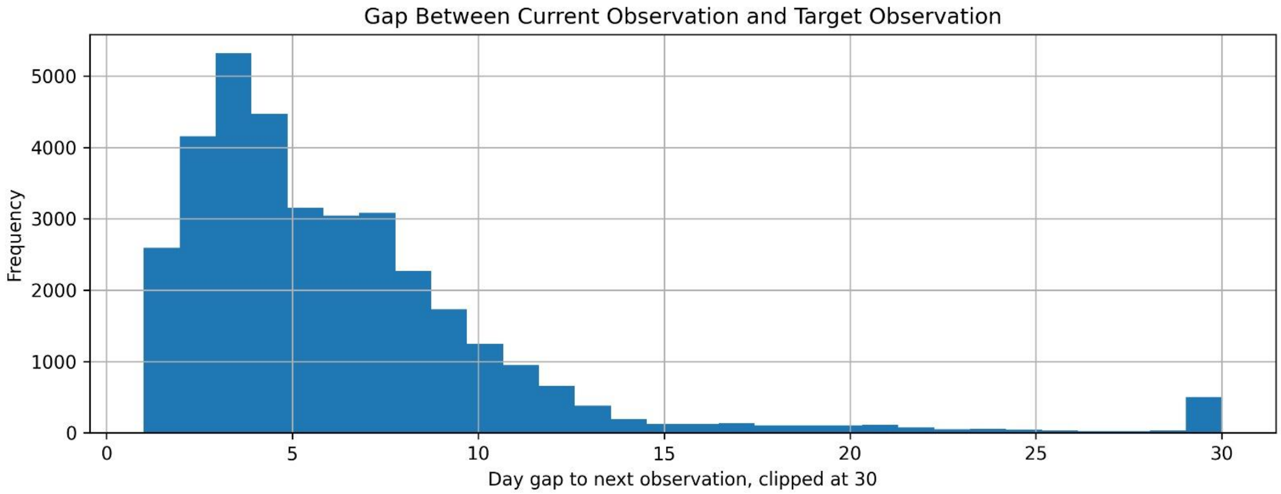

Figure 4: Distribution of day gaps between the current observation and target observation. Most test cases have a gap of one day.

10.4. Correlation Heatmap

Correlation analysis was used to understand relationships between numerical features.

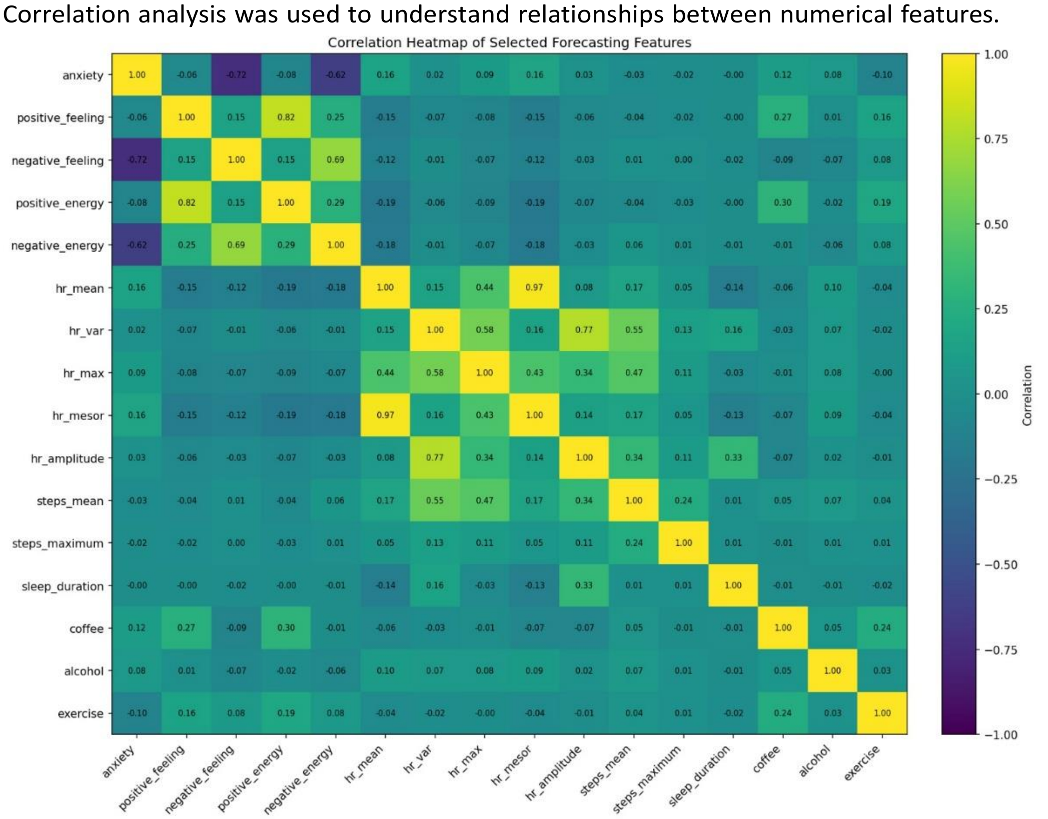

Figure 5: Correlation heatmap of numerical features. Related physiological and mental-health variables show strong correlations.

Some important high correlations observed in the raw analysis include:

Table 10: Strongly Correlated Feature Pairs

Feature 1	**Feature 2**	**Correlation**
hr_mean	hr_mesor	0.973
bandpower_0_0005_0_0001hz	bandpower_0_00005_0_00001hz	-0.902
agoraphobia	social_phobia	0.900
anxiety_control	body_shape	0.876
fear_negative_evaluation	social_avoidance_distress	0.862
agoraphobia	interoceptive_fear	0.857
hr_amplitude	bandpower_0_00005_0_00001hz	0.841
social_phobia	interoceptive_fear	0.833
positive_feeling	positive_energy	0.818
sleep_onset_time	sleep_out_time	0.805
steps_maximum	steps_hvar_mean	0.789
hr_var	hr_amplitude	0.774
generalized_anxiety	state_anxiety	0.695
social_phobia	state_anxiety	0.693

Correlation helps identify redundant variables and related feature groups. However, it does not imply causation.

10.5. ACF and PACF Plots

Autocorrelation and partial autocorrelation plots were used to understand lag dependency in anxiety.

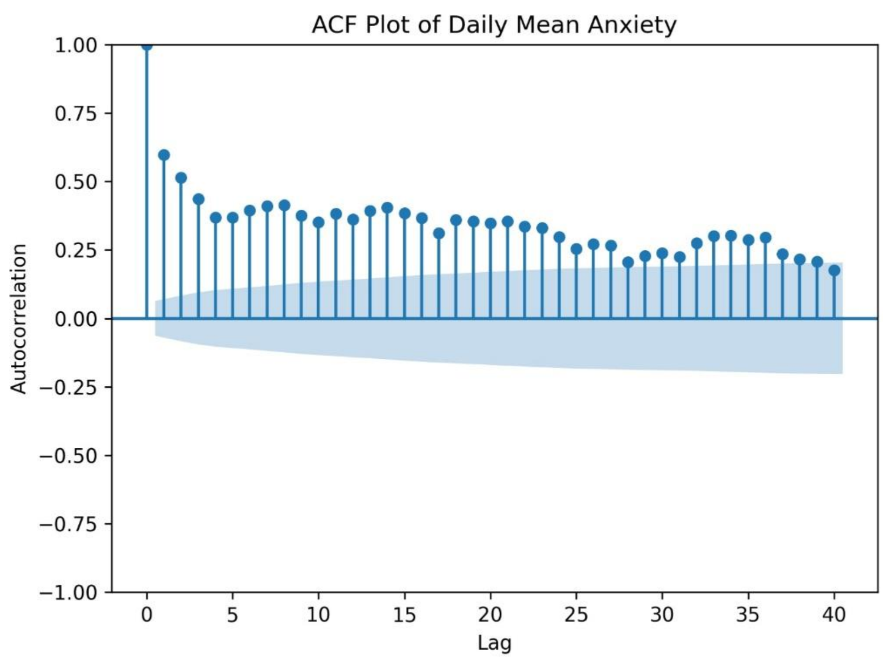

Figure 6: ACF plot for anxiety. Autocorrelation helps identify whether previous anxiety values are related

to future anxiety.

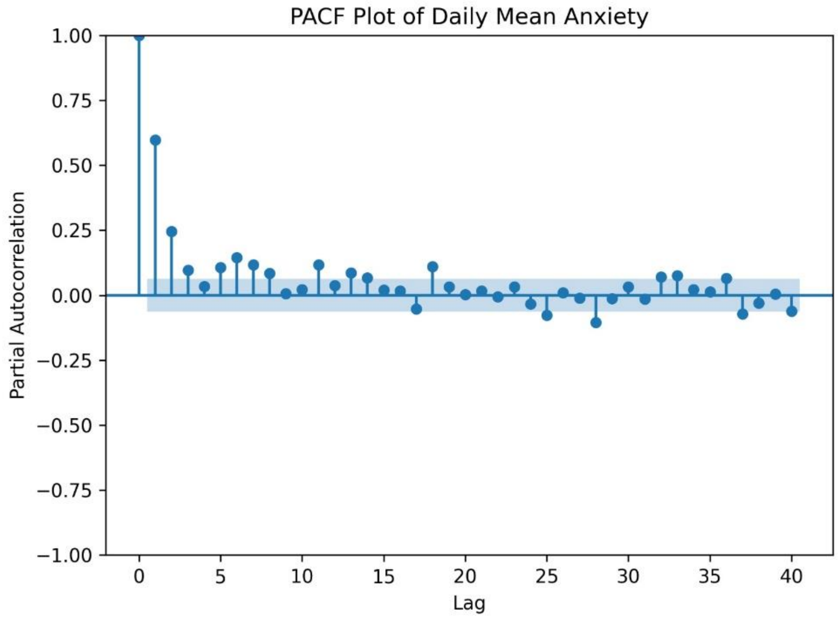

Figure 7: PACF plot for anxiety. PACF helps identify direct lag relationships after removing intermediate lag effects.

The strong performance of Naive Persistence and SMA-10 supports the idea that recent anxiety history is important.

10.6. Stationarity Check

For statistical time-series models such as ARIMA and SARIMA, stationarity is important. A stationary series has relatively stable mean and variance over time.

The Augmented Dickey-Fuller test checks whether a time series is stationary.

The hypotheses are:

$$H_0 : \text{the series is non-stationary}$$

$$H_1 : \text{the series is stationary}$$

If the p-value is less than 0.05, the series is usually treated as stationary.

In this project, stationarity analysis is mainly useful for statistical models such as AR, MA, ARMA, ARIMA, SARIMA, and SARIMAX. Since the final dataset is a multi-participant panel dataset rather than one long regular time series, stationarity testing is used as supporting analysis, not as the only model-selection criterion.

10.7. Differencing

If a time series is non-stationary, first-order differencing can be applied:

$$\Delta y_t = y_t - y_{t-1}$$

Differencing focuses on change rather than absolute level. ARIMA includes this idea through the integration parameter *d*.

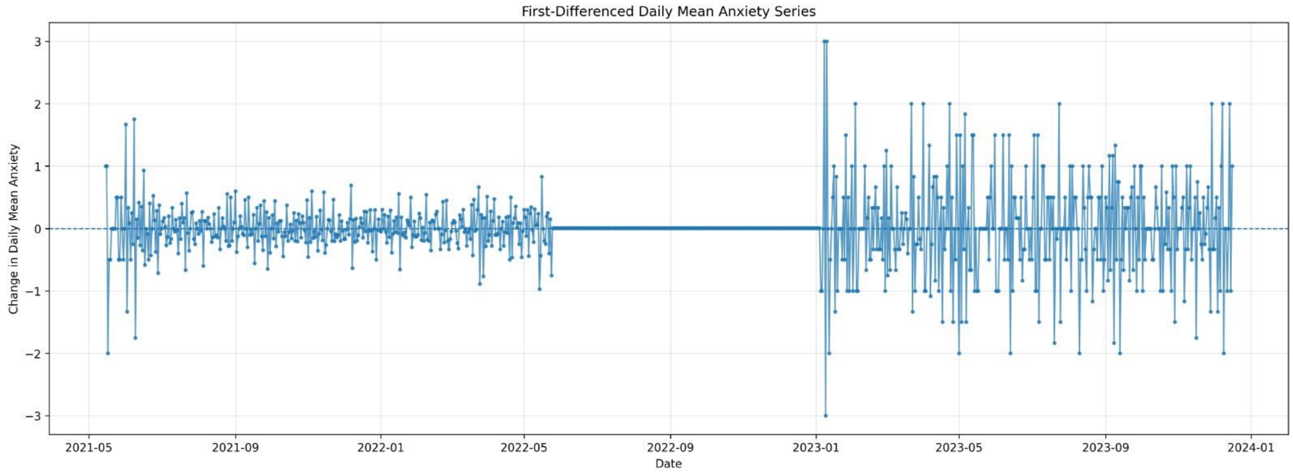

Figure 8: Differenced anxiety series. Differencing is used to reduce non-stationarity for ARIMA-type models.

11. Seasonality Consideration

The project is short-term daily or near-daily forecasting. The main model input uses the previous 10 observations.

For SARIMA and SARIMAX, weekly seasonality was considered:

$$m = 7$$

This corresponds to a 7-observation cycle. Since most records are daily, this approximately represents weekly seasonality.

Monthly or yearly seasonality was not used because:

1. individual participant series are not long enough,
2. the records are not perfectly regular for all participants,
3. the forecasting goal is short-term, not yearly,
4. large seasonal periods would make SARIMA/SARIMAX unstable.

Therefore, the main seasonality considered was weekly seasonality.

12. Models Compared

We compared baseline models, statistical time-series models, machine-learning models, and deep-learning models. Time-series forecasting can be approached from different methodological traditions. Statistical models (ARIMA, SARIMAX, etc.) capture temporal structure like autocorrelation and seasonality but rely primarily on the target's own history. Machine learning models (XGBoost, Random Forest, etc.) handle multivariate, nonlinear interactions across many features simultaneously. Deep learning models (LSTM, GRU) are designed for sequence data and long-range dependencies. Since this project uses a multi-participant panel dataset with 16 features across a 10-observation window, no single approach is guaranteed to dominate. Therefore, all three families were compared systematically.

12.1. Baseline Models

Baseline models are simple reference models. They help check whether complex models are actually useful.

12.1.1. Naive Persistence

Naive Persistence assumes:

$$\hat{y}_{t+1} = y_t$$

It predicts that the next anxiety score will be equal to the current anxiety score.

12.1.2. SMA-10

SMA-10 predicts using the average of the last 10 anxiety values:

$$\hat{y}_{t+1} = \frac{1}{10}$$

It smooths noise and captures recent average anxiety level.

12.1.3. Linear Regression Baseline

Linear Regression uses lagged features to learn a straight-line relationship between the recent history and the future anxiety score.

12.2. Statistical Time-Series Models

12.2.1. AR

The Autoregressive model uses previous target values:

$$y_t = c + \phi_1 y_{t-1} + \cdot \cdot \cdot + \phi_p y_{t-p} + \epsilon_t$$

12.2.2. MA

The Moving Average model uses previous forecast errors:

$$y_t = \mu + \epsilon_t + \vartheta_1 \epsilon_{t-1} + \cdot \cdot \cdot + \vartheta_q \epsilon_{t-q}$$

12.2.3. ARMA

ARMA combines AR and MA components:

$$ARMA(p, q)$$

It uses both past values and past errors.

12.2.4. *ARIMA*

ARIMA adds differencing to ARMA:

$$ARIMA(p, d, q)$$

The parameter d controls how many times the series is differenced.

12.2.5. *SARIMA*

SARIMA adds seasonal terms to ARIMA:

$$SARIMA(p, d, q)(P, D, Q)_m$$

In this project, $m = 7$ was used for weekly seasonality.

12.2.6. *SARIMAX*

SARIMAX extends SARIMA by adding exogenous variables. This allows external features such as sleep, steps, or heart-rate variables to support forecasting.

12.3. Machine-Learning Models

Machine-learning models use lag-window features to learn supervised forecasting patterns.

12.3.1. *Random Forest*

Random Forest averages multiple decision trees. It reduces overfitting compared to a single tree and captures nonlinear patterns.

12.3.2. *XGBoost*

XGBoost is a gradient-boosted tree model. It builds trees sequentially, where each new tree tries to correct errors made by previous trees. It was the best continuous forecasting model in our final results.

12.3.3. *Gradient Boosting*

Gradient Boosting is another boosting method that combines weak learners step-by-step to reduce prediction error.

12.3.4. *Decision Tree*

A Decision Tree creates if-then rules based on features. It is interpretable but can overfit if not controlled.

12.3.5. *K-Nearest Neighbors*

KNN predicts using the most similar past examples in feature space. It is simple but sensitive to scaling and noise.

12.3.6. Support Vector Regression

SVR uses margin-based regression and can model nonlinear relationships using kernels.

12.3.7. Ridge Regression

Ridge Regression uses L2 regularization:

$$+ \lambda \sum_{j=1}^{p} \beta_j^2$$

This shrinks coefficients and reduces overfitting.

12.3.8. Lasso Regression

Lasso Regression uses L1 regularization:

$$\lambda \sum_{j=1}^{p} |\beta_j|$$

This can shrink some coefficients to zero and perform feature selection.

12.3.9. Linear Quantile Regression

Linear Quantile Regression predicts specific quantiles of the target distribution. In this project, it was used to estimate lower, median, and upper risk boundaries.

12.4. Deep-Learning Models

12.4.1. LSTM

LSTM is a recurrent neural network designed to learn temporal dependencies using memory gates.

12.4.2. GRU

GRU is a simpler recurrent model than LSTM. It has fewer gates and can train faster while still capturing temporal dependencies.
Deep-learning models were included because they are commonly used for sequence forecasting. However, they did not outperform XGBoost in this dataset.

12.4.3. Why LSTM and GRU Were Not Ensembled?

Ensembling multiple LSTM or GRU models was not attempted for two main reasons. First, both deep learning models performed worse individually than XGBoost, with GRU achieving RMSE ≈ 0.6007 and LSTM showing even higher error. Ensembling weak models rarely produces a strong result. Second, the dataset contains only 3458 supervised sequences, which is relatively small for deep learning. Training even a single LSTM or GRU required careful regularization to avoid overfitting. An ensemble would multiply the data requirement and computational cost without a clear path to outperforming XGBoost. Therefore, deep learning ensembling is left as potential future work if a larger dataset becomes available.

13. Final Experimental Workflow

The final workflow was:

1. Load 0017_jang_ts.tsv.
2. Sort records by participant and date.
3. Select final wearable, activity, sleep, lifestyle, and self-report features.
4. Create the next-observed anxiety target using participant-wise shift.
5. Calculate day gap between the input date and the target date.
6. Create 10-observation rolling windows.
7. Split each participant chronologically into 70% train and 30% test.
8. Train baseline, statistical, machine-learning, and deep-learning models.
9. Evaluate continuous forecasting using MAE, RMSE, and R^2.
10. Evaluate high-risk warning using precision, recall, and F1-score.
11. Evaluate the trained predictions again on only true next-day samples where day_gap = 1.

14. Evaluation Metrics

14.1. Mean Absolute Error

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

MAE measures the average absolute prediction error.

14.2. Root Mean Squared Error

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

RMSE penalizes large errors more strongly than MAE.

14.3. R^2 Score

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

R^2 measures how much variance in the target is explained by the model.

14.4. High-Risk Classification Metrics

High-risk anxiety is defined as:

$$\text{High-Risk} = \begin{cases} 1 & \text{if anxiety} \geq 2 \\ 0 & \text{if anxiety} < 2 \end{cases}$$

For high-risk warning, we evaluate:

- Accuracy,
- Precision,
- Recall,
- F1-score.

For a warning system, recall is especially important because missing high-risk cases is undesirable.

15. Main Results: Next-Observation Forecasting

The main experiment used all 1050 test samples. The target was next-observed anxiety.

15.1. Top Models by RMSE

Table 11: Top Models on Full Next-Observation Test Set by RMSE

Model	N	MAE	RMSE	R^2	Default Recall	Tuned F1
XGBoost	1050	0.3581	0.5838	0.6270	0.3564	0.6467
Lasso Regression	1050	0.3644	0.5872	0.6227	0.3515	0.6542
SMA-10	1050	0.3368	0.5938	0.6141	0.3762	0.6407
Gradient Boosting	1050	0.3706	0.5945	0.6132	0.3812	0.6376
Random Forest	1050	0.3729	0.5962	0.6110	0.3020	0.6477
GRU	1050	0.3651	0.6007	0.6051	0.3663	0.6427
Linear Quantile Regression	1050	0.3239	0.6020	0.6033	0.3515	0.6465
Ridge Regression	1050	0.3847	0.6068	0.5969	0.3762	0.6490
ARMA	1050	0.3906	0.6073	0.5963	0.3564	0.6346
MA	1050	0.3850	0.6097	0.5932	0.3564	0.6254

15.2. Main Finding

XGBoost was the best continuous forecasting model:

$$RMSE \approx 0.584$$

$$R^2 \approx 0.627$$

This means that XGBoost explained approximately 62.7% of the variation in next-observed anxiety on the test set.

16. Complete Model Result Tables

This section summarizes the main numerical results from the model comparison.

16.1. Main Leaderboard by RMSE

Table 12: Main Leaderboard by RMSE: Full Next-Observation Test Set

Model	MAE	RMSE	R^2	Model Family
XGBoost	0.3581	0.5838	0.6270	Machine Learning
Lasso Regression	0.3644	0.5872	0.6227	Machine Learning
SMA-10	0.3368	0.5938	0.6141	Baseline
Gradient Boosting	0.3706	0.5945	0.6132	Machine Learning
Random Forest	0.3729	0.5962	0.6110	Machine Learning
GRU	0.3651	0.6007	0.6051	Deep Learning
Linear Quantile Regression	0.3239	0.6020	0.6033	Machine Learning
Ridge Regression	0.3847	0.6068	0.5969	Machine Learning
ARMA	0.3906	0.6073	0.5963	Statistical
MA	0.3850	0.6097	0.5932	Statistical

16.2. Next-Observation vs True Next-Day Performance

Table 13: XGBoost Performance: Next-Observation vs True Next-Day

Evaluation Setting	N	MAE	RMSE	R^2
All next-observation test samples	1050	0.3581	0.5838	0.6270
Only true next-day test samples	861	0.3564	0.5839	0.6043

The RMSE remains almost unchanged. This proves that XGBoost's performance is still strong when evaluated only on true next-day cases.

17. True Next-Day Subset Validation

To check whether the model works for actual next-day forecasting, the same predictions were evaluated only on test samples where:

$$\text{day gap} = 1$$

There were 861 true next-day samples.

17.1. Top Models on True Next-Day Samples

Table 14: Top Models on True Next-Day Subset by RMSE

Model	N	MAE	RMSE	R^2	Default Recall	Tuned F1
XGBoost	861	0.3564	0.5839	0.6043	0.3312	0.6154
Lasso Regression	861	0.3639	0.5868	0.6004	0.3248	0.6296
Gradient Boosting	861	0.3684	0.5941	0.5903	0.3567	0.6058
Random Forest	861	0.3671	0.5956	0.5882	0.2611	0.6159
SMA-10	861	0.3372	0.5963	0.5872	0.3503	0.6193
GRU	861	0.3606	0.5988	0.5838	0.3312	0.6058
Linear Quantile Regression	861	0.3220	0.6008	0.5810	0.3185	0.6284
Ridge Regression	861	0.3858	0.6069	0.5725	0.3376	0.6189
ARMA	861	0.3889	0.6081	0.5708	0.3185	0.6167
MA	861	0.3840	0.6102	0.5678	0.3185	0.5983

17.2. True Next-Day Finding

XGBoost remained the best continuous forecasting model:

$$RMSE \approx 0.584$$

$$R^2 \approx 0.604$$

The RMSE is almost unchanged compared to the full next-observation test set. This shows that the model is not only working because of mixed irregular next-observation gaps. It remains effective on true next-day samples.

18. High-Risk Alerting Results

The high-risk threshold was:

$$\text{anxiety} \geq 2$$

At the default threshold, some models were conservative and missed high-risk cases. Therefore, tuned thresholds were also tested.

18.1. Tuned Alert Results on True Next-Day Subset

Table 15: Top Models on True Next-Day Subset by Tuned Alert F1

Model	RMSE	R^2	Tuned Threshold	Tuned Recall	Tuned F1
ARIMA	0.6187	0.5557	1.24	0.6752	0.6424
SARIMAX	0.6755	0.4703	1.28	0.6433	0.6372
Lasso Regression	0.5868	0.6004	1.24	0.6497	0.6296
Linear Quantile Regression	0.6008	0.5810	1.18	0.6624	0.6284
SARIMA	0.6465	0.5149	1.24	0.6752	0.6272
Support Vector Regression	0.6147	0.5614	1.16	0.6624	0.6228
SMA-10	0.5963	0.5872	1.12	0.6943	0.6193
Ridge Regression	0.6069	0.5725	1.16	0.6879	0.6189
ARMA	0.6081	0.5708	1.26	0.6815	0.6167
Random Forest	0.5956	0.5882	1.28	0.5669	0.6159
XGBoost	0.5839	0.6043	1.16–1.28	0.3312–0.5669	0.6154
GRU	0.5988	0.5838	1.16–1.28	0.3312–0.5669	0.6058
Gradient Boosting	0.5941	0.5903	1.16–1.28	0.3567–0.5669	0.6058
MA	0.6102	0.5678	1.16–1.28	0.3185–0.5669	0.5983

18.2. Interpretation

The best model for continuous score prediction was XGBoost. However, the best model for tuned alert F1 on the true next-day subset was ARIMA.

This shows:

> The best score predictor is not always the best warning model.

For practical warning, calibrated thresholds are important. A model may predict the score accurately on average but still require a lower threshold to catch high-risk cases.

19. Graphical Results

19.1. Dataset and Time-Series Plots

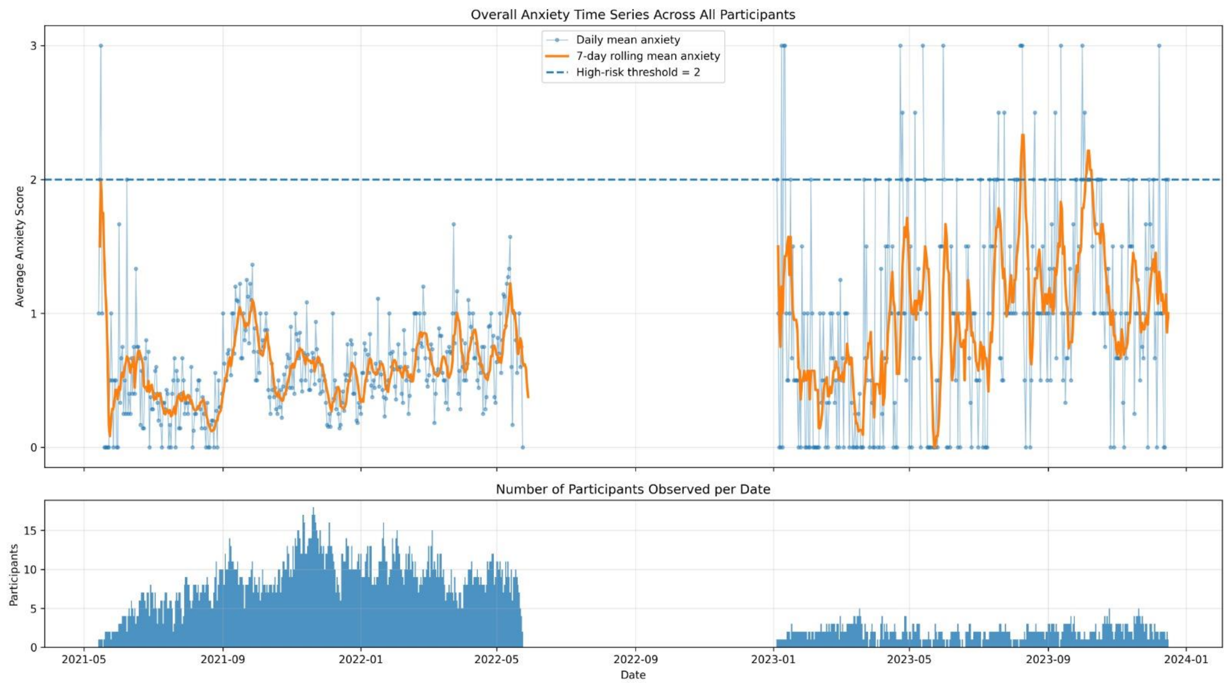

Figure 9: Main anxiety time-series visualization.

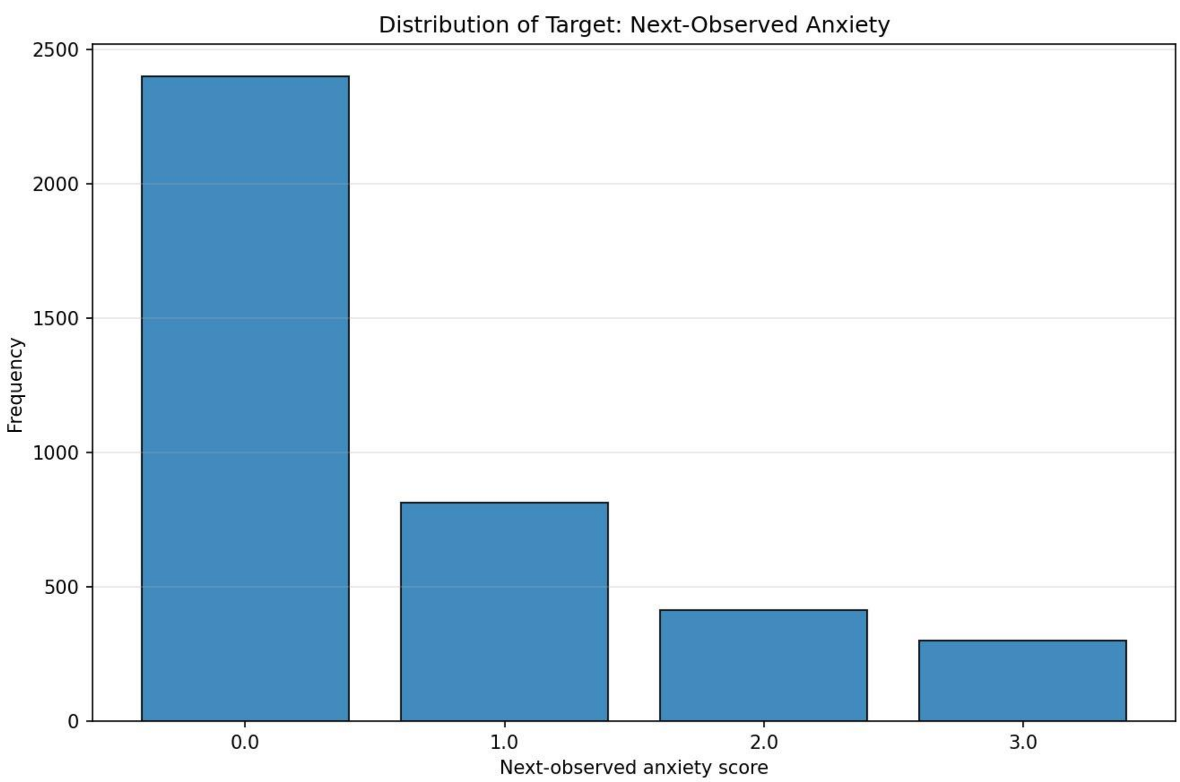

Figure 10: Target distribution of next-observed anxiety.

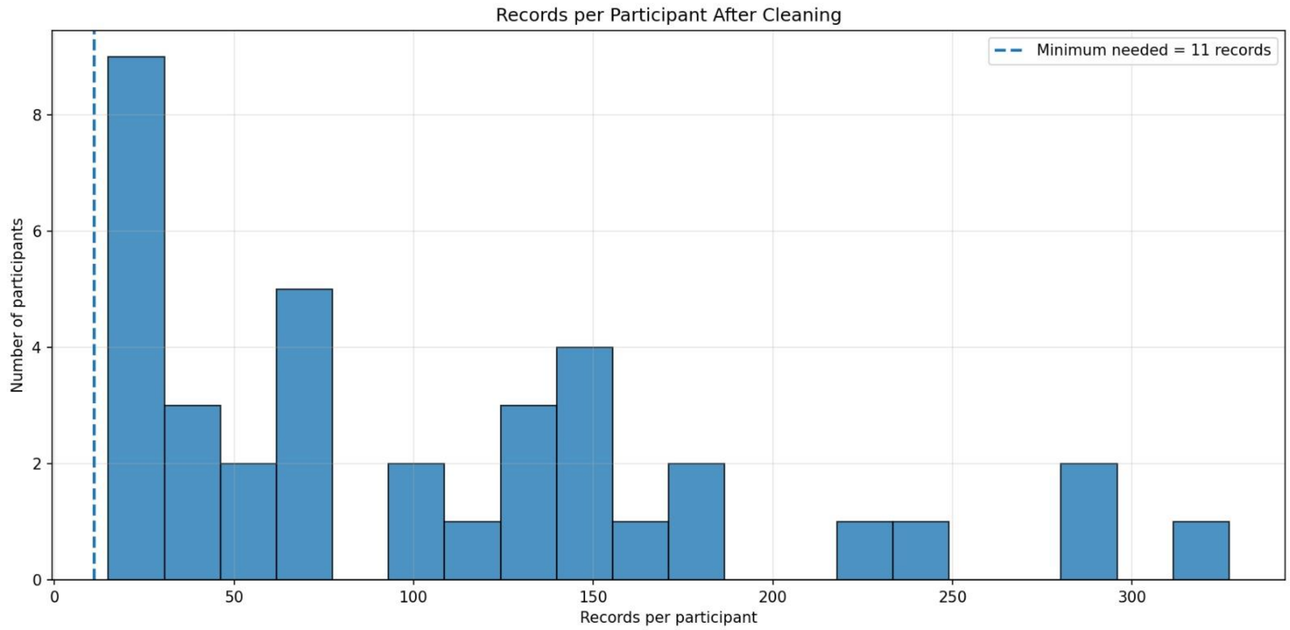

Figure 11: Records per participant after cleaning and filtering.

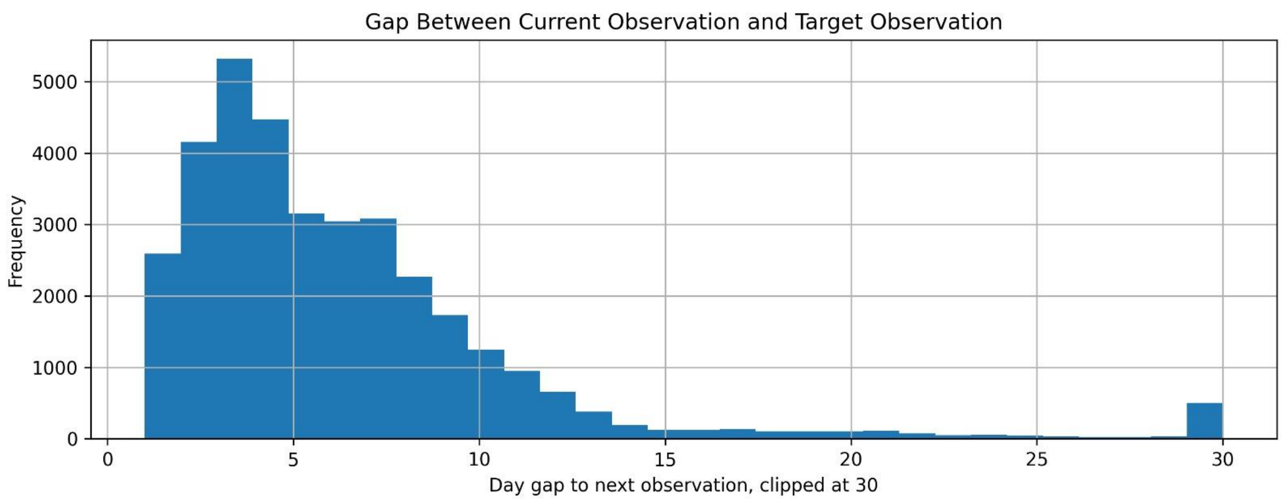

Figure 12: Day-gap distribution showing that most test examples are true next-day cases.

19.2. All-Model Comparison

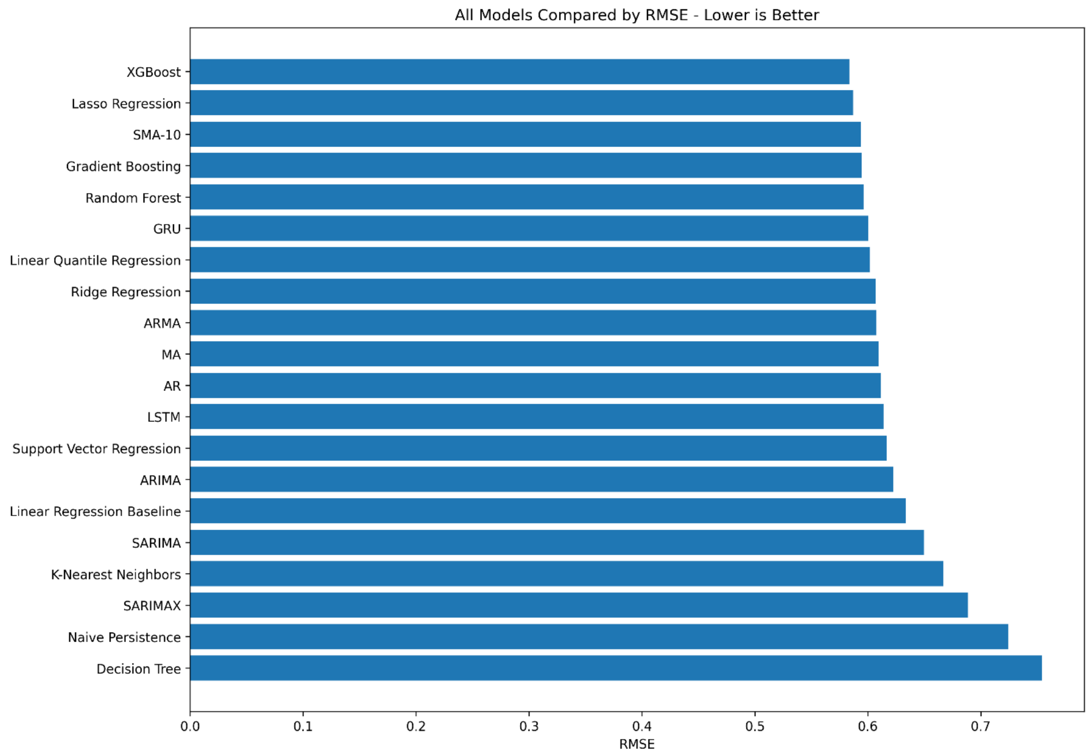

Figure 14: Comparison of all models by RMSE. Lower RMSE is better.

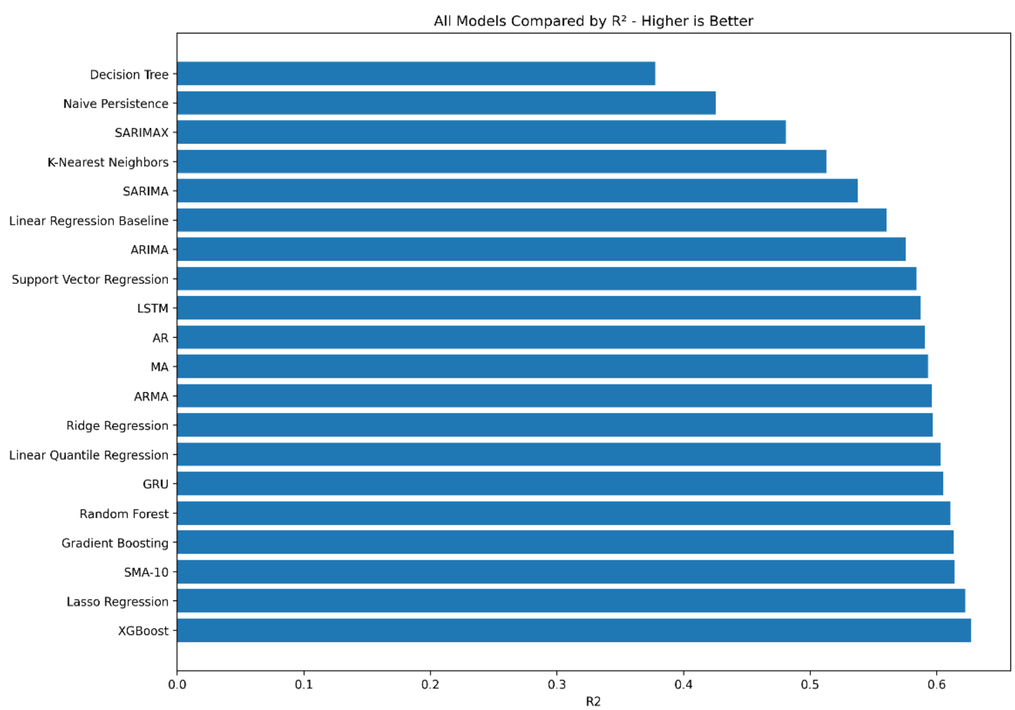

Figure 15: Comparison of all models by R^2. Higher R^2 is better.

19.3. Best Model and Individual Model Plots

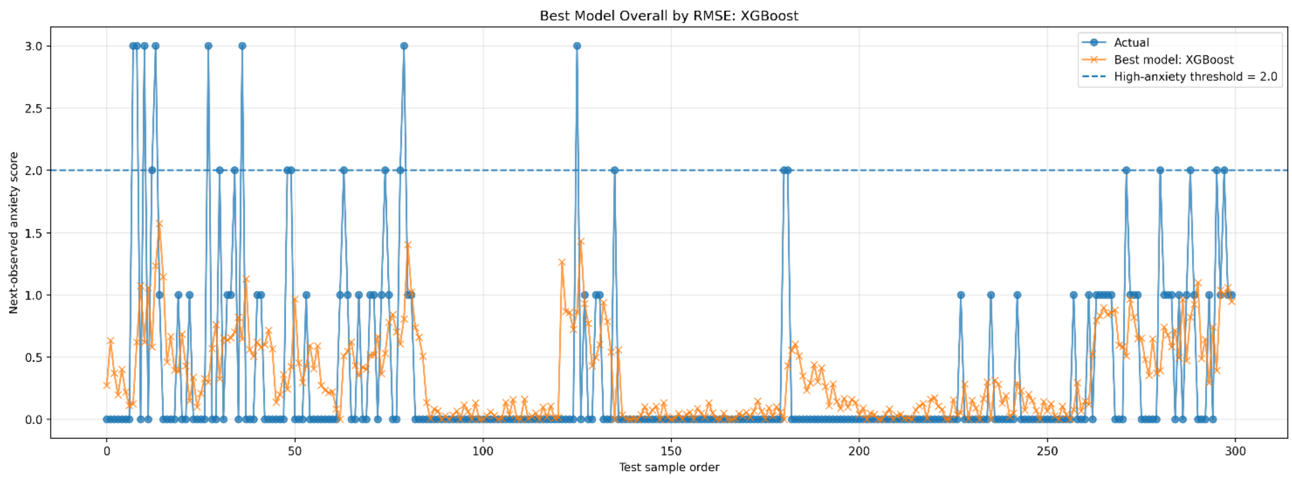

Figure 16: Best model actual vs predicted plot. XGBoost captures broad anxiety trends but still misses some sudden spikes.

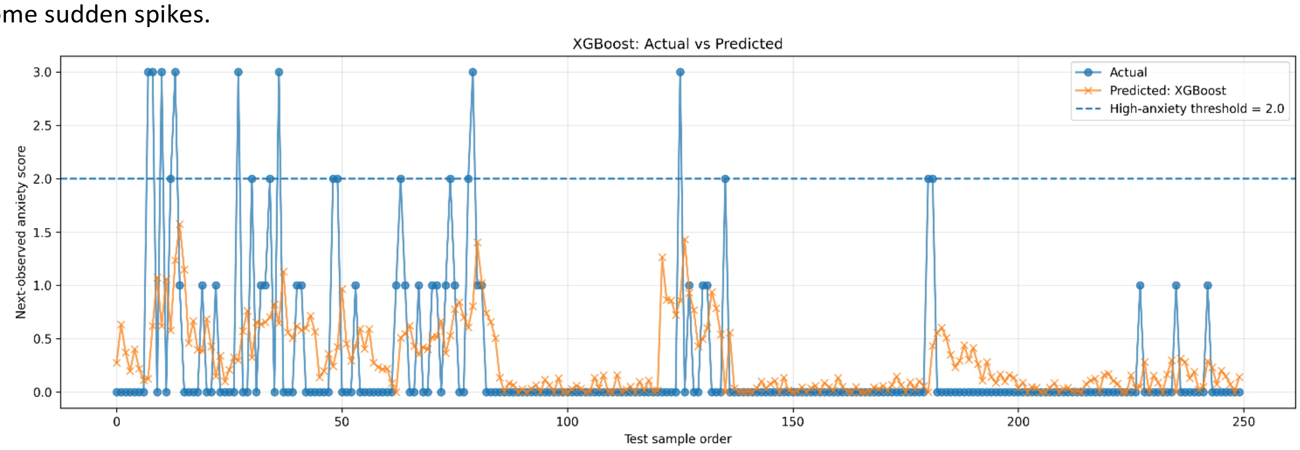

Figure 17: XGBoost actual vs predicted plot.

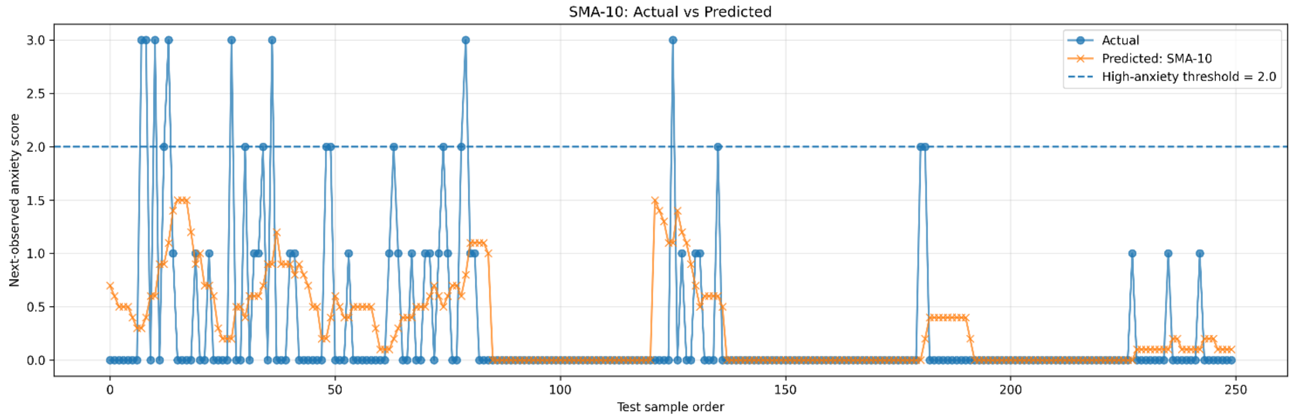

Figure 18: SMA-10 actual vs predicted plot.

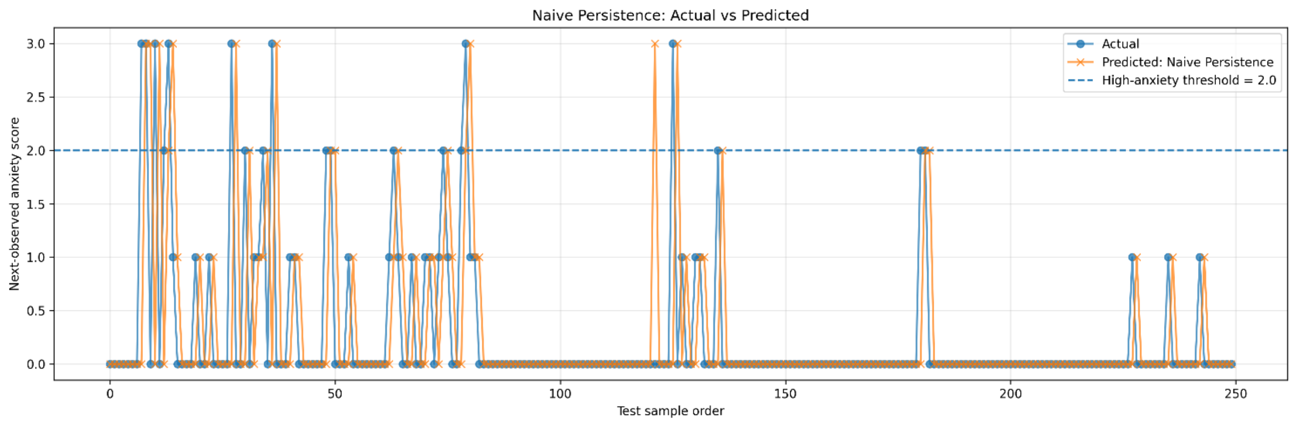

Figure 19: Naive Persistence actual vs predicted plot.

19.4. Quantile Regression Bounds

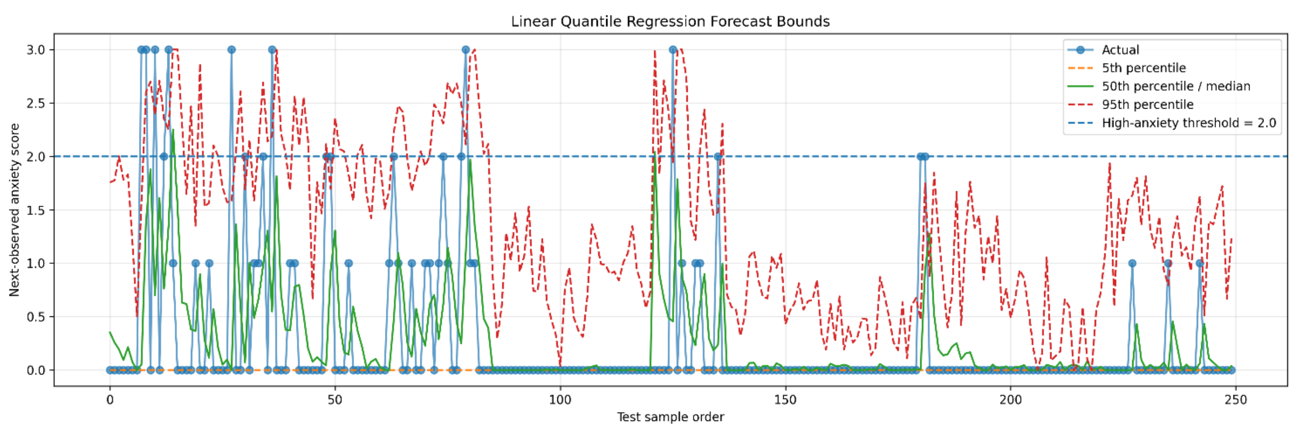

Figure 20: Linear Quantile Regression forecast bounds. The 95th percentile can be interpreted as a conservative upper-risk boundary.

19.5. ACF, PACF, and Differencing Plots

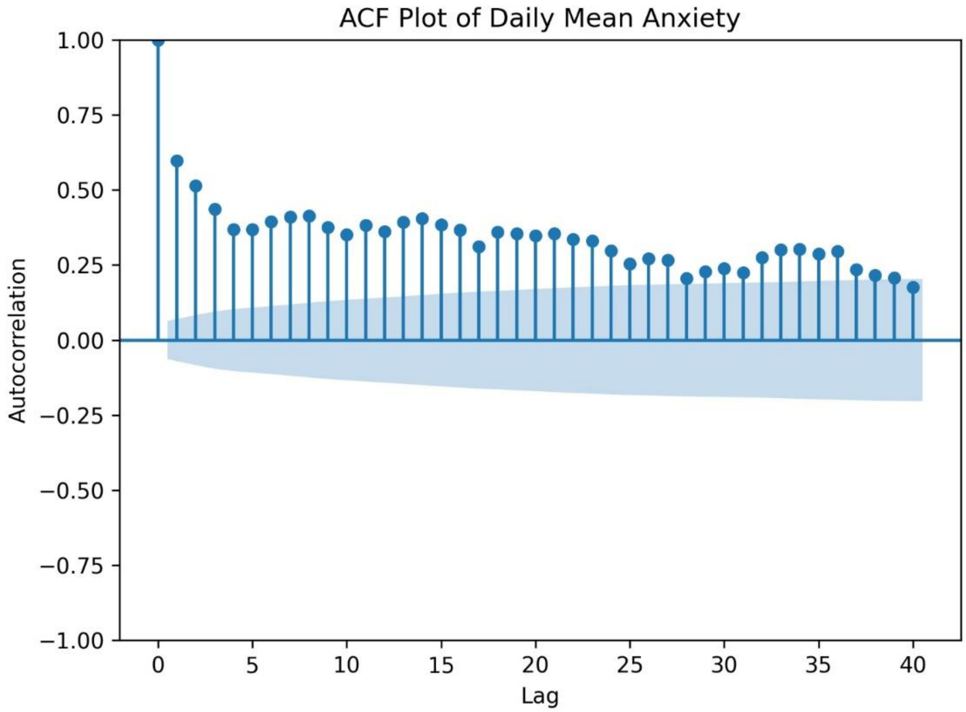

Figure 21: ACF plot for anxiety.

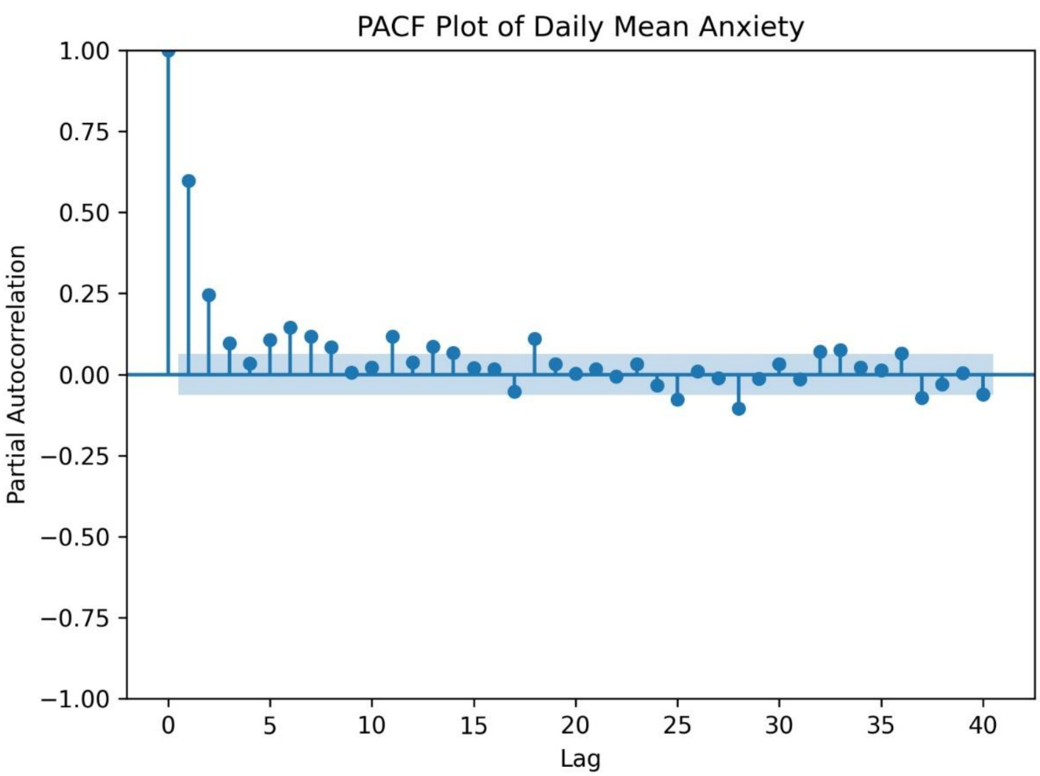

Figure 22: PACF plot for anxiety.

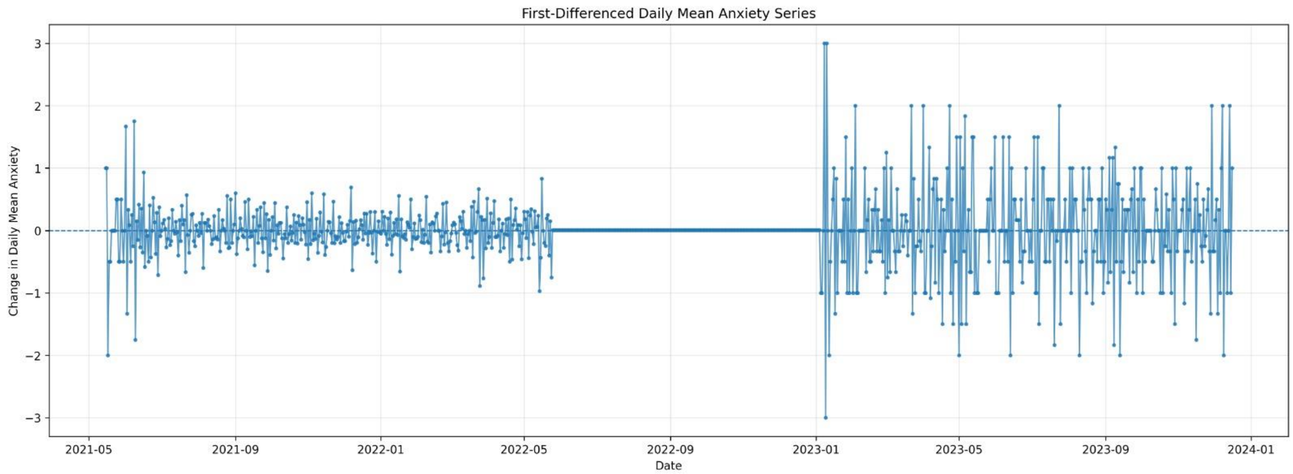

Figure 23: Differenced anxiety series.

19.6. Deep Learning Training Curves

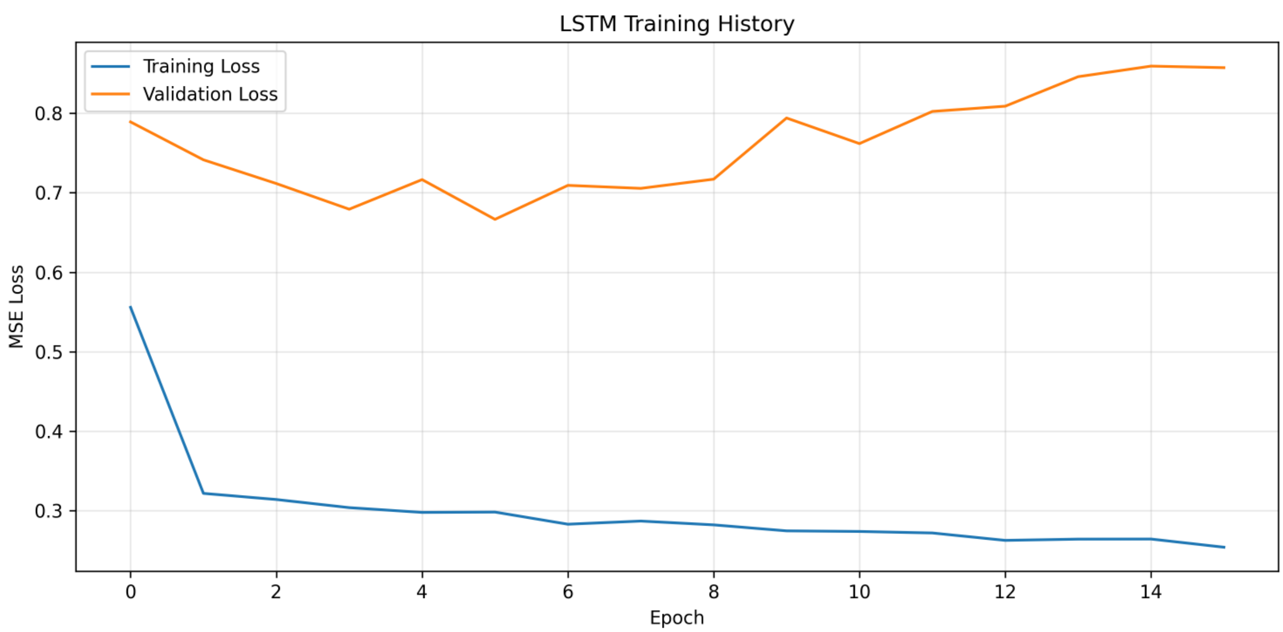

Figure 24: LSTM training and validation loss.

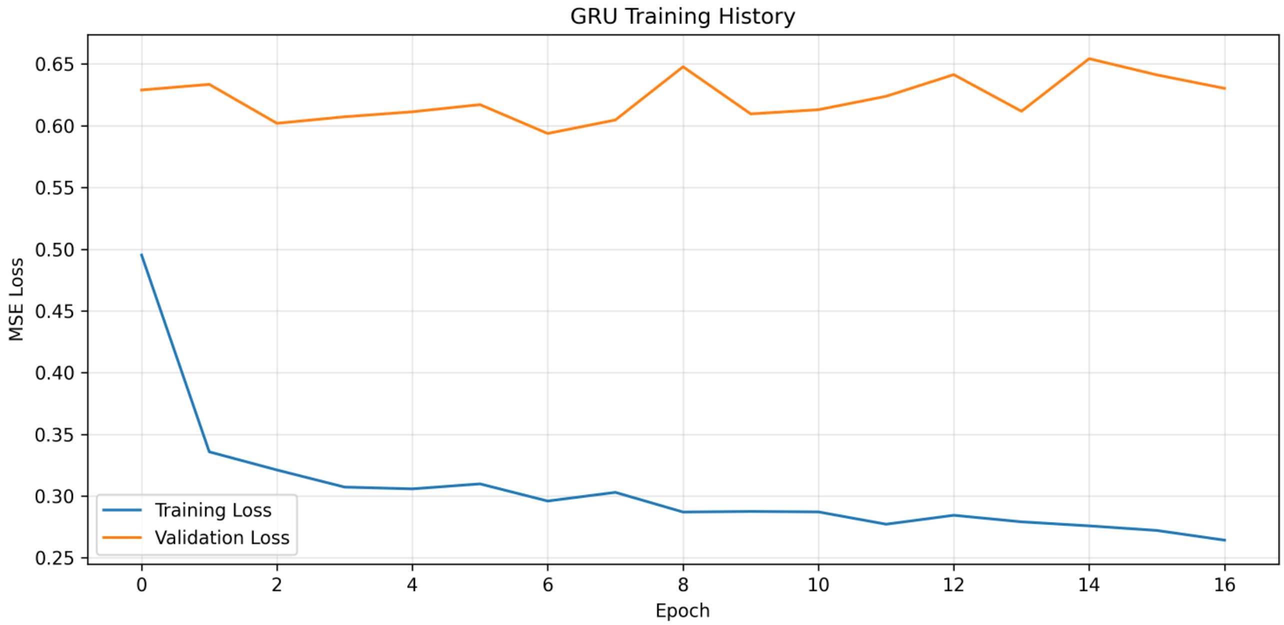

Figure 25: GRU training and validation loss.

20. Why XGBoost Performed Best

XGBoost performed best because the final dataset is not a single smooth univariate time series. It is a multi-participant, multivariate, panel-style time-series dataset.

Statistical models such as ARIMA and SARIMAX mainly rely on temporal structure. They work best when the data is regular, stable, and has clear autocorrelation or seasonality.

In contrast, XGBoost can learn nonlinear patterns from many lagged features together:

- previous anxiety,
- mood variables,
- sleep duration,
- step features,
- heart-rate features,
- coffee, alcohol, and exercise variables.

This makes XGBoost more suitable for this task.

The result does not mean that SARIMAX is useless. SARIMAX remains a useful statistical baseline, especially for alerting after threshold tuning. But for continuous anxiety-score prediction, XGBoost performed best.

21. Machine Learning vs Statistical Time-Series Models

The final results show a difference between statistical time-series models and machine-learning models.

21.1. Statistical Time-Series View

Statistical models such as ARIMA and SARIMA mainly ask:

> How much does past anxiety explain future anxiety?

They are useful because anxiety has short-term persistence.

21.2. Machine-Learning View

Machine-learning models ask:

> How do recent anxiety, mood, sleep, heart rate, activity, and behavior jointly predict future anxiety?

This is more suitable for this project because mental-health risk is not only a clock-based pattern. It can depend on multiple signals together.

21.3. Final Interpretation

The final interpretation is:

> Short-term anxiety forecasting is a cross-signal forecasting problem. Past anxiety is important, but recent wearable, sleep, activity, behavioral, and self-report context can also provide useful information.

22. Data Leakage and Validity Analysis

22.1. Target Leakage Check

A major concern in forecasting projects is data leakage. Data leakage happens when the model uses information that would not be available at prediction time.

In our final dataset, the target is future anxiety:

$$\text{current and past features} \rightarrow \text{next observed anxiety}$$

The target is not directly calculated from current heart-rate, sleep, step, or mood features. Therefore, there is no obvious direct formula-based target leakage.

22.2. Use of Current Anxiety

The model uses current anxiety as part of the input window. This is valid if the prediction is made after the current self-report/check-in is available.

The practical interpretation is:

> After today's observation is available, including today's anxiety and other features, the model forecasts the next observed anxiety score.

This is not leakage because the target is future anxiety.

22.3. What the Model Should Not Claim

Because self-report variables are included, the model should not be described as wearable-only prediction. The correct description is:

Forecasting using wearable, sleep, activity, behavioral, and self-report history.

23. Feature Count and Model Complexity

The final input contains:

$$10 \text{ observations} \times 16 \text{ features} = 160 \text{ flattened features}$$

This is acceptable, but the dataset is not extremely large. Therefore, using too many features may lead to overfitting.

A reduced feature set could be tested in future work:

$$\{\text{anxiety}, \text{sleep\_duration}, \text{hr\_mean}, \text{steps\_mean}, \text{positive\_feeling}, \text{negative\_feeling}, \text{exercise}, \text{coffee}\}$$

This would make the model simpler and easier to interpret.

Recommended feature ablation experiments are:

1. anxiety-only history,
2. mood/self-report history only,
3. wearable and behavior features only,
4. reduced interpretable feature set,
5. full feature set.

These experiments would help determine whether the model mainly depends on previous anxiety or whether wearable and behavioral features add extra predictive power.

24. Comparison with general_ema.csv

The general_ema.csv dataset was useful because it had a direct stress target. However, its forecasting horizon was less clear because only about 5.45% of test cases were exactly next-day.

Table 16: Final Comparison: 0017_jang_ts.tsv vs general_ema.csv

Criterion	0017_jang_ts.tsv	general_ema.csv
Target	Anxiety	Stress
Forecast task	Next-observed / mostly next-day	Next-observed stress
Users after filtering	28	204
Supervised sequences	3458	33058
True next-day test cases	82%	about 5.45%
Best model	XGBoost	XGBoost
Best RMSE	about 0.584	about 0.825
Best R^2	about 0.627	about 0.404
Feature richness	Wearable + sleep + activity + self-report	EMA/self-report mainly
Final suitability	Better for short-term forecasting	Better only if direct stress target is mandatory

Therefore, 0017_jang_ts.tsv was selected as the final dataset.

25. Discussion

25.1. Main Findings

The main findings are:

1. XGBoost was the best continuous anxiety-score forecasting model.
2. XGBoost remained strong on true next-day samples.
3. SMA-10 and Naive Persistence performed strongly, showing that recent anxiety history is important.
4. Deep-learning models did not outperform XGBoost.
5. Statistical time-series models were useful baselines, but they were not the best continuous predictors.
6. High-risk warning requires threshold calibration.
7. The best continuous predictor is not always the best high-risk alert model.

25.2. Why Baselines Were Strong

Anxiety has temporal persistence. If a person is anxious today, they may still be at risk tomorrow. Therefore, simple models like Naive Persistence and SMA-10 are meaningful baselines.

The fact that complex models must compete with these baselines makes the evaluation more honest.

25.3. Why Deep Learning Did Not Win

LSTM and GRU models are powerful, but they usually require larger datasets to outperform tree-based models. In this project, the number of users and sequences was limited. Therefore, XGBoost performed better because it works well on medium-sized tabular lag-window data.

25.4. Prediction vs Early Warning

The final project has two related but different goals:

1. predict the next anxiety score accurately,
2. identify high-risk anxiety cases early.

XGBoost was best for the first goal. For the second goal, calibrated thresholds and persistence-based logic are important.

26. Limitations

The project has the following limitations:

1. The target is anxiety, not direct burnout or clinical stress.
2. The system is not a diagnostic tool.
3. The main target is next-observed anxiety, although 82% of test samples are true next-day.

4. The model is evaluated using within-user chronological split, not leave-user-out validation.
5. Sudden anxiety spikes are difficult to predict.
6. Self-report variables are included, so the model is not wearable-only.
7. Deep-learning performance is limited by dataset size.
8. The final dataset has only 28 users after filtering.
9. ADF, ACF, and PACF are more naturally suited to single regular time series than multi-user panel data.
10. Some exact plots or secondary diagnostics depend on the generated output files from the final pipeline.

27. Future Work

Future improvements include:

1. Train and test the full pipeline only on strict true next-day pairs.
2. Compare 70/30 and 80/20 chronological splits.
3. Perform leave-user-out validation to test unseen-user generalization.
4. Run feature ablation experiments.
5. Tune XGBoost, LSTM, and GRU hyperparameters more deeply.
6. Build a hybrid early-warning system combining XGBoost, persistence, and quantile upper-bound alerts.
7. Use personalized models for participants with enough records.

28. Conclusion

This project developed a short-term mental-health risk forecasting system using 0017 jang ts.tsv. The target was next-observed anxiety, and 82% of the test samples were true next-day cases. Therefore, the task can be interpreted as mostly next-day anxiety forecasting.

The dataset was converted into a supervised time-series forecasting format using a 10-observation input window. Baseline models, statistical time-series models, machine-learning models, and deep-learning models were compared.

XGBoost achieved the best continuous forecasting performance:

$$RMSE \approx 0.584$$

$$R^2 \approx 0.627$$

On the true next-day subset, XGBoost remained strong:

$$RMSE \approx 0.584$$

$$R^2 \approx 0.604$$

This confirms that the model performs well not only on mixed next-observation data, but also on actual next-day cases.

The project also found that high-risk warning is different from score prediction. XGBoost is the best continuous predictor, but calibrated thresholds and persistence-based logic are important for high-risk anxiety alerts.

The final conclusion is:

> Short-term anxiety risk is moderately predictable using recent wearable, sleep, activity, behavioral, and self-report history. XGBoost is the strongest continuous forecasting model, while calibrated alert rules are important for practical warning.